Sabari Vishnu Jayanthan Jaikrishnan

Staff AI / ML Engineer | Staff Data Scientist

Ontario, Canada | Canadian Permanent Resident | Authorized to work in Canada without sponsorship

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Professional Summary

Staff-level AI/ML engineer and data scientist with 7 years of experience building production-grade RAG, agentic AI, and computer vision systems. Technical lead for multi-site AI/data platform delivery across global operations, translating complex operational data into decision-ready insights and taking systems from prototype to measurable production impact.

Career Highlights

- Served as technical lead for a multi-site AI/data platform, driving production output from 30K to 32K parts/day, and delivered ML risk models mitigating ~$12M in annual losses.
- Architected and shipped production RAG systems — LangChain, pgvector, hybrid vector+BM25 retrieval, human-feedback tuning — cutting enterprise knowledge-discovery time from 40+ hours to minutes.
- Built agentic AI systems using MCP tool servers with controlled, retrieval-grounded tool execution, reducing hallucination and unsafe-action risk while recommending optimized machine parameters in production.

Technical Skills

- **Product Analytics / Data Science:** Deep Learning, Computer Vision, Predictive ML, Gradient Boosting, Random Forest, Time Series Forecasting, Classification, Regression, Clustering, A/B Testing, Anomaly Detection, Risk Detection
- **Data / Analytics:** Apache Kafka, PostgreSQL, pgvector, MongoDB, MySQL, Dataiku, Power BI, Tableau, TIBCO Spotfire, Matplotlib
- **Programming:** Python, SQL, JavaScript, React, Node.js, Flask, FastAPI, REST APIs, Microservices
- **ML Frameworks:** TensorFlow, PyTorch, pandas, NumPy, scikit-learn
- **GenAI:** Generative AI, Agentic AI, RAG pipelines, LangChain, prompt engineering, hallucination mitigation, OpenAI text-embedding-3, hybrid search (vector + BM25), semantic chunking, vector search, Claude API, OpenAI API, MCP
- **Cloud / MLOps:** AWS, AWS Lambda, Docker, Kubernetes, Jenkins CI/CD, OpenVINO, edge inference (Intel Movidius NCS), model deployment, production monitoring, Git

Leadership & Delivery

Owned end-to-end architecture for cross-functional AI/data initiatives — from problem framing through production adoption — partnering with operations, engineering, and leadership stakeholders, and influencing leadership decisions on AI/data infrastructure investment based on platform outcomes.

Professional Experience

Staff Data Scientist | Western Digital (WD)

January 2021 – Present

- Architected a cloud-native RAG platform (AWS Lambda, Claude API, LangChain, pgvector) that indexed thousands of documents across SharePoint, wikis, and engineering reports using structure-aware chunking and hybrid vector+BM25 retrieval. Reached **100+ users**, with SME-feedback-driven accuracy tracking via Power BI, and cut enterprise knowledge-discovery time from **40+ hours to minutes**.
- Designed the architecture for a real-time AI-driven decision-optimization platform, built in Python with MCP tool servers and pgvector for controlled tool execution, then containerized it with Docker and Kubernetes across multiple production lines to **recommend optimized machine parameters within seconds** while reducing hallucination and unsafe-action risk.
- Served as technical lead for multi-site AI/data platform delivery across 3 global operations, standardizing reusable ML workflows, KPI tracking, deployment patterns, and monitoring practices that drove a production **output increase from 30K to 32K parts per day**.
- Built a real-time computer vision system using TensorFlow SSD and HLS streaming, deploying inference on an Intel Movidius Neural Compute Stick with OpenVINO 2024 at the edge and a Python/React/Node.js stack, to monitor workforce-to-equipment utilization, **improving operational efficiency by 16%** and contributing to a World Economic Forum Global Lighthouse recognition.
- Productionized a gradient boosting risk-detection pipeline, orchestrated through Jenkins CI/CD, to identify hidden defect patterns tied to **~$12M in annual losses**, shifting decision-making from reactive analysis to proactive risk mitigation.
- Mentored 3-5 junior and intermediate engineers on ML practices including model evaluation, testing, and versioning through code and design reviews, driving consistent production ML deployment across teams.

Systems Engineer | Infosys Pvt. Ltd. (Client: U.S.-Based Automotive Enterprise)

June 2016 – June 2018

- Led migration of a legacy billing system to Oracle SOA BPEL on Oracle Fusion Middleware while embedded on a U.S. enterprise client account, integrating SQL, XML, and REST APIs for production-grade data access.
- Built reusable data pipelines and warehouse integration workflows, improving data consistency across transactional systems and downstream reporting.
- Worked directly with client business and IT stakeholders, translating requirements into BPEL process designs and supporting delivery scope and timelines.

Education

Master of Science in Information and Data Science

King Mongkut's University of Technology North Bangkok, Thailand – 2020 | CGPA: 4.0 / 4.0

Bachelor of Technology in Electronics and Communication Engineering

SRM University, India – 2016 | CGPA: 8.5 / 10